

Beyond the Checklist: Rethinking Trustworthiness in AI System

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Abstract—Trustworthiness is often operationalized through a defined set of requirements — e.g., robustness, reliability, transparency, explainability, fairness, accountability, privacy — that systems are expected to meet to be considered trustworthy. With the advent of Artificial Intelligence (AI), the concept of trustworthiness has evolved significantly, expanding beyond merely technical dimensions to encompass ethical and legal nuances. Despite these broadened considerations, the predominant checklist-oriented approach, in which machines are deemed trustworthy upon fulfilling a predetermined set of criteria, remains prevalent. Although this approach might appear natural and beneficial, it risks oversimplifying its inherent complexity.

This paper explores this approach with a dual purpose. First, it argues that the conceptual ambiguity currently surrounding trustworthiness crucially impedes effective interdisciplinary communication and risks promoting superficial compliance — i.e., “ethics-washing” — where systems might be labeled trustworthy primarily due to their technical performance, without factually respecting their ethical implications and broader societal impacts, or vice versa. Second, it acknowledges that trustworthiness cannot be merely an intrinsic system attribute but is fundamentally context-dependent, user-relative, and dynamically evolving within human-machine interactions - thus becoming a dynamic concept shaped by these elements. In this light, it is also important not to neglect the notion of *perceived trustworthiness*, which is introduced to highlight the role of subjective evaluations shaped by contextual and individual factors. These aspects collectively influence trust in AI-powered systems.

By integrating technical rigor, ethical awareness, and user-centric evaluations, this paper seeks to refine conceptual clarity, particularly within the domain of requirements engineering.

I. INTRODUCTION

Can we really “trust” artificial intelligence (AI) - or may we merely “rely” on it? As AI systems become ever more embedded and adopted in our professional, social and private lives, the question of ethical alignment and trust in these systems is crucial [1]. First, we need to account the fact that the term trustworthiness is often used ambiguously - sometimes carelessly - to describe a requirement to support fruitful human-computer interaction.

Yet the initial question demands an initial clarification between the concepts of reliance and trustworthiness. In the philosophical orthodoxy, accounts of trustworthiness often distinguish it from mere reliability by grounding it in moral or normative features. One influential tradition explains this distinction in psychological terms, particularly through the

concept of goodwill. According to Baier [2], trustworthy individuals reliably fulfill commitments out of goodwill toward the person who trusts them. This moral orientation—rather than mere performance—accounts for why trust can be betrayed, whereas reliance cannot. Potter [3] builds on this by framing trustworthiness as a virtue: a stable disposition to respond appropriately to being trusted, shaped by one’s identity and moral character, including virtues like justice and compassion. On these views, trustworthiness involves not just what a person does, but why they do it — their motives and moral traits are central. So, trustworthiness brings with it a moral dimension in the interaction.

In contrast, Hawley [4], [5] offers a commitment-based view that strips away the psychological and moral load. On her account, trustworthiness is a matter of avoiding unfulfilled commitments. A trustworthy person is one who does not take on more than they can handle and is diligent in meeting the commitments they do accept. Importantly, this view is morally neutral: fulfilling commitments does not require goodwill or virtuous character. It is a “negative” account, in that someone can be trustworthy simply by refusing to make commitments they might not keep. Yet trustworthiness still differs from mere reliability, because it is tied to the presence (or absence) of commitments—one can be reliable in action without ever having committed to those actions, and thus not qualify as trustworthy.

Together, these approaches offer two contrasting visions: one rooted in moral psychology and virtue, the other in cautious and consistent adherence to commitments, regardless of internal motivations.

In accordance with the contribution by Blanco (2025: [6]), that is aligned with our positionality on the issue, we argue that reliance refers to the assessment of functional, operative and technical effectiveness: a system is relied upon when it behaves consistently as expected, and in accordance with the purposes for which it is engineered, thus delivering predictable and technically correct outcomes in line with its design. This notion is rooted in what can be called computational reliabilism [6], and it is connected to a performance-based and more traditional evaluation of technologies’ functions. However, the increasing autonomy and unpredictability of modern AI-powered systems - particularly due to their expanding decision-making capabil-

ities, aimed at reproducing human-like reasoning - call for a shift in the way we interpret their impact on society. In this sense, the very notion of reliance no longer captures the kind of relationship that human agents are establishing with these systems, especially in complex interaction environments (i.e., high-stakes application domains).

This is the point where trustworthiness comes into play: not as an articulation of the notion of reliance (much less as a synonym for it), but as a more complex, layered concept that captures the relational dimension of human-AI interaction. Following this distinction, we argue that trustworthiness cannot simply be reduced to the technical success of the system. Rather, even its technical dimension must be understood and conceptualized as oriented toward supporting positive interactions with human agents. In this sense, we differentiate between reliance, which refers to the system's isolated functional capabilities, and technical trustworthiness, that concerns whether the system performs in a way that enables and sustains appropriate and quality interactions.

Thus, even the technical layer of trustworthiness is not neutral, because it needs to be shaped by an orientation and an attention toward human agents and the impact of the system. This primary distinction lays the basis for understanding the further layers of complexity that arise when we consider the conceptual articulation of trustworthiness, which will be discussed in the sections that follow.

Paper roadmap. Section II overviews the state of the art in trustworthiness covering both the technical and ethical perspective. Section III discusses the conceptual confusion around trustworthiness and explains how this can lead to the risk of ethics washing. Section IV discusses the impact of the above considerations in the realms of requirement engineering and quality assurance. Section V discusses a path towards a more mature conceptualization of trustworthiness and a precise conceptual framework.

II. TRUSTWORTHY AI: FROM TECHNICAL TO ETHICAL

Trustworthiness often refers to rigorous system design, i.e., designing systems so that they behave safely and guarantee security or quality aspects. When focusing on AI systems, trustworthiness is likely reduced to guaranteeing that systems have been designed and built to actually do what they are supposed to do [7]. However, modern AI-enabled systems challenge existing techniques due to the inherent complexity of formulating correct specifications and devising appropriate adequacy metrics to evaluate their effectiveness [8]. Given this initial technical complexity that needs to be assessed in a more objective-oriented way, trustworthiness also entails a subjective dimension, that is related to perceived trustworthiness (that will be detailed in Section V) and to the relation that it has with the notion of trust.

In fact, trust is precisely the dual of trustworthiness [7]. Trust concerns the acceptability of systems from the point of view of humans [9], then calling for a more subjective point of view: humans can easily reject algorithmic prescriptions that “authoritatively” make decisions or predictions. For instance,

transparency and explainability are certainly qualities that can help establish a trust relationship between humans and machines.

The ISO/IEC TS 5723:2022 standard defines trustworthiness as the ability to meet stakeholders' expectations in a verifiable way¹. This definition is interesting since it introduces the stakeholders' expectations, having the context of use implicit, and the fact that trustworthiness should be verifiable, asking then for measurability and demonstrability by means of objective evidence.

The ISO/IEC TS 5723:2022 standard also mentions accountability, accuracy, authenticity, availability, controllability, integrity, privacy, quality, reliability, resilience, robustness, safety, security, transparency, and usability as characteristics of trustworthiness. Also the NIST Trustworthy & Responsible Artificial Intelligence Resource Center (AIRC) identifies the characteristics trustworthy AI systems should have: valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair with harmful bias managed². These characteristics have been used to construct a taxonomy of trustworthiness for artificial intelligence that includes 150 properties, each of which constitutes a dimension of one of the characteristics [10].

Overall, we observe an agreement on the multidisciplinary nature of trustworthiness. For example, Trustworthiness for AI in Defense [11] identifies different dimensions of trustworthiness, namely military, technical, and legal, where not only technical factors but also normative values play a role in the human-machine cooperation. Even in the critical nature of military missions and the potential consequences of AI system failures or misuse, the word trustworthiness is a key concept. In this particular context, the ambition is to inspire confidence among the various stakeholders that AI technologies operate predictably, securely, and ethically in complex and dynamic environments. Here, we can observe two aspects in particular: the role of trustworthiness to inspire confidence to stakeholders, opening to a subjective belief, which in other sources we see more in the sphere of trust [7], and the aspect of ethics. Concerning the subjectivity, the Industrial Internet Consortium defines trustworthiness as the “degree of confidence one has that the system performs as expected. Characteristics include safety, security, privacy, reliability and resilience in the face of environmental disturbances, human errors, system faults and attacks” [12]. Concerning ethical aspects, also the AI Act [13] clearly states that trustworthy AI should be (i) lawful - respecting all applicable laws and regulations, (ii) ethical - respecting ethical principles and values, (iii) robust - both from a technical perspective while taking into account its social environment³. The taxonomy in [10] also spans over law, ethics, and technical aspects.

Concluding, the available definitions and descriptions of trustworthiness for AI, AI systems or, more in general,

¹<https://www.iso.org/obp/ui/en/#iso:std:iso-iec:ts:5723:ed-1:v1:en>

²<https://airc.nist.gov/airmf-resources/airmf/3-sec-characteristics/>

³<https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>

for autonomous systems are not crystal clear, refer to sub-dimensions or characteristics that are only in partial agreement, include sometimes also ethical and moral judgment, and do not even agree on its objectivity or subjectivity nature. This justifies why we need a conceptual engineering of trustworthiness - i.e., a critical and systematic analysis of the concept - as discussed in the next section.

As for the philosophical point of view, especially in epistemology and action theory, trustworthiness is often defined in functional terms, with no essential ethical content.

Howley [4] and Simion and Kelp [14] suggest that trustworthiness can be understood primarily in terms of competence, reliability, and responsiveness, and that it does not necessarily entail moral virtues. From this perspective, something or someone can be considered trustworthy in an instrumental, pragmatic, or predictive sense, meaning: “This system will reliably perform as expected.” This definition focuses on the system’s ability to meet expectations consistently, without an inherent moral or ethical dimension.

In contrast, when institutions such as the European Union refer to “trustworthy AI,” they are not adopting the narrower, philosophical conception of trustworthiness. Instead, the term carries significant normative content — it implies that the system is not only reliable but also ethical, encompassing qualities such as fairness, transparency, accountability, and non-discrimination. Here, the phrase “Trust me” transcends mere functional reliability and implicitly suggests: “Trust me, I am ethical, fair, and aligned with your values.”

Is this the same concept at all? We can fairly argue not necessarily — as we have demonstrated. To put it roughly, in philosophy, trustworthiness is often considered value - neutral or ethically thin, focusing primarily on qualities like reliability and competence without requiring any particular ethical or moral framework. In contrast, within the realm of AI regulation, trustworthiness becomes ethically thick: it is imbued with normative content and is often used to legitimize and frame systems as morally acceptable, ethical, or aligned with societal values such as fairness, transparency, and accountability. This shift highlights the difference between the technical and ethical senses of trustworthiness, with the latter being a concept laden with moral expectations, as explored in earlier sections.

III. CONCEPTUAL CONFUSION - ETHICS-WASHING RISK

The differing approaches to trustworthiness — both across disciplines such as philosophy, computer science, and policy, and even within the same discipline — give rise to three major issues that are not independent but deeply interconnected:

- 1) Conceptual confusion: The term “trustworthy” is used across various disciplines—such as philosophy, computer science, economics, and policy—but often with very different meanings. This leads to misunderstandings and miscommunication across fields, undermining productive dialogue and collaboration [15], [16].
- 2) Ethics-washing: Technical systems are sometimes branded as “trustworthy” based solely on their robustness, accuracy, or reliability, even when they lack essential

ethical properties such as fairness, transparency, or respect for fundamental rights. This allows developers and institutions to project ethical credibility without making substantive ethical commitments or addressing real-world harms.

- 3) Normative ambiguity in frameworks and guidelines: Some existing conceptual models (e.g., [17]) and policy guidelines (e.g., [18]) define a system’s ethical quality in terms of values like beneficence, non-maleficence, fairness, or societal and environmental well-being. However, these values are not neutral or universally defined—what counts as “doing good” or being “fair” varies significantly across social, cultural, and political contexts. This issue arises at the intersection of conceptual confusion and ethics-washing: ambiguity in ethical terminology enables inconsistent interpretations, while ethics-washing exploits this vagueness to mask a lack of meaningful ethical oversight or engagement with affected communities.

We must either accept that “trustworthy AI” is a distinct, ethically-loaded concept that diverges from traditional philosophical usage, or we should advocate for greater conceptual clarity by differentiating between (i) Functional trustworthiness, as found in philosophy and engineering, and (ii) Normative trustworthiness, as emphasized in AI governance.

IV. CONFUSION FROM QUALITY ASSURANCE AND REQUIREMENT ENGINEERING

This section is aimed to unveil further confusion that arises within requirement engineering and quality assurance. Sometimes trustworthiness is considered as one of the “-ilities” to be considered, and sometimes trustworthiness seems to be the final end.

Trustworthiness as a mean. In technical frameworks and standards [18], [19], trustworthiness is often treated as one quality among others that contributes to ensuring effective and positive human-AI interaction, alongside others e.g., robustness, explainability, safety, privacy, fairness, etc. In this sense, trustworthiness appears as one requirement within a list, something that can be embedded and assessed in parallel with other system’s properties - that, when combined, can prove the *objective* quality of the system.

Trustworthiness as an end. However, in different context (e.g., AI ethics and governance), trustworthiness is positioned at a higher level of abstraction, and treated as a holistic property of the system - that is achieved only if multiple and combined underlying requirements are satisfied. Consistently, in this view a system is considered trustworthy because it is defined as robust, safe, secure, fair, explainable, etc.: according to this, trustworthiness is not a single requirement among others, but rather a property that depends on the fulfillment of several conditions.

This creates a first and crucial structural ambiguity: understanding trustworthiness as a requirement vs. trustworthiness

as a qualitative-outcome of multiple underlying requirements is critically different, and complicates both the design and the evaluation of AI-powered systems. In the context of Requirement Engineering, this ambiguity would need to be resolved, as it remains unclear whether trustworthiness is the end and the goal to be achieved, or one of the several means to achieve a broader notion of system's quality. We support the idea that the first step in enabling the resolution of this ambiguity must be moved on the path toward a conceptual clarity that is able to integrate the precision required by philosophical perspectives with the rigor needed to overcome technical challenges.

V. TRUSTWORTHINESS: A PATH TO CONCEPTUAL CLARITY

As it also recognized within the literature from different disciplines, the concept of trustworthy AI needs some refinement [20], [21], [22]. Indeed, to clarify the concept of trustworthiness within the context of AI system design, it is necessary to move beyond surface-level definitions and toward a precise conceptual framework. As demonstrated throughout this paper, the ambiguity surrounding trustworthiness stems, at least in part, from the fact that the term is used and investigated across distinct domains, each with its own assumptions and expectations, and sometimes subject to difficult interdisciplinary communication with each other. To support more efficient and clear requirement elicitation processes, we propose distinguishing between three interrelated, yet analytically and operatively distinct, interpretations of trustworthiness in AI-powered systems: i.e., technical, ethical and perceived trustworthiness. This section of the paper aims to support conceptual clarity between these notions and to clarify how they are closely related to the operationalization of trustworthiness in the requirement engineering domain.

Technical trustworthiness. It refers to the properties that make the system able to fulfill its functions and to complete its tasks under expected and unexpected conditions [7]. Technical requirements pertain to all the properties that are typically engineered during the development phase, and that can be often quantified or measured through empirical means. These attributes are strongly linked with the system's performance capacity. Indeed, qualities like robustness, resilience and safety relate to how well the system can handle unforeseen and critical situations, while properties like explainability are more referred to the system's interpretability during the interaction with the user. From this perspective, technical trustworthiness is linked with the performance integrity and operational correctness, and is tied to design-phase methods.

Ethical trustworthiness. It introduces a layer of expectation regarding the system's consistency with fundamental rights and moral and societal values. Indeed, ethical requirements extend beyond the mere technical functionality and also involve commitments about what the system *ought* to do in order to be perceived trustworthy. These qualities can include, for example, fairness, privacy and accountability,

which are more nuanced features that specifically address how the system interacts with human values, moral boundaries, rights, as well as with social norms and expectations [23]. In this regard, it is fundamental to highlight that the operationalization of ethical trustworthiness through predefined requirements is a crucial yet difficult challenge: many of the qualities involved in this layer are inherently nuanced and, in some ways, contingent - and they all need a proper contextualization to be assessed. For instance, the quality of beneficence [24] is not a universally accepted principle: what counts "benevolent" in one context, or from one perspective, may be perceived as non-benevolent from another. This inner ambiguity makes it extremely critical to translate ethical expectations into formal requirements without oversimplifying them.

Perceived trustworthiness. It emphasizes the user's subjective expectations, beliefs, and experience with the system's - and the alignment between them and the system's actual behavior. Also, this layer is nuanced, and, most importantly, it is neither entirely dependent to the system properties nor completely under the user's control. Actually, it emerges from various contextual cues, as well as from the interaction's dynamics, their socio-cultural framing and the user's preferences. In this view, perceived trustworthiness is a mediator between objective, fulfilled requirements and experienced human-AI relationship. Moreover, it is fundamental to note that it is one of the elements that shape the user's general sense of *trust* in the system, together with other influencing factors [25].

Such conceptual differentiation is crucial in the Requirement Engineering domain. First, it allows stakeholders (from development to evaluation teams) to disambiguate what kind of trustworthiness is involved: are we trying to guarantee that the system will behave consistently and correctly (*technical*)? That it will respect and be aligned with rights, norms, values and principles (*ethical*)? Or are we focusing on the fact that it will be experienced as trustworthy, credible, engaging by users (*perceived*)? The clarification of these categories shapes the requirement elicitation process, and contributes to sustain the fact that trustworthiness needs to be treated as a multi-dimensional, complex construct, in which each dimension raises different needs for requirements, demands different validation processes, and involves different stakeholders.

Moreover, we sustain the idea that trustworthiness is not a static property that can be universally attributed to a system once, forever and for all - and in this section we are going to delve into this dynamic lens. First, trustworthiness is a situated and contingent quality, shaped by the environmental (*space*), temporal (*time*) conditions, as well as by subjective factors and other social, cultural and dispositional elements that have been previously discussed in more detail. In particular, the spatial dimension refers to the specific socio-technical context in which an AI-powered system is deployed: in this sense, trustworthiness emerges as an output from the alignment between

the system's functioning and the infrastructural expectations that are embedded in the environment of use. For instance, a conversational AI used for medical advice may be seen as trustworthy in a hospital due to its adherence to clinical protocols and institutional oversight. However, if the same AI is deployed in a home setting without institutional support or oversight, it may be perceived as unreliable or risky due to the lack of accountability and formal control mechanisms. The change in space—from a controlled hospital environment to an informal home setting—dramatically alters its trustworthiness.

The temporal dimension, by contrast, reveals the very dynamic nature of trustworthiness over time. Indeed, trustworthiness is a temporal-dependent and -sustained relationship. Indeed, the trustworthiness of a new AI system may be high at launch, when it is rigorously tested and controlled. Over time, as the system becomes outdated, fails to adapt to new data, or experiences unforeseen issues, its trustworthiness can decline. This shift over time highlights the dynamic nature of trustworthiness, as it is influenced by the system's ongoing performance, updates, and maintenance. This reveals that trustworthiness is not just verifiable at design time, but must be maintained, updated, and negotiated across the lifecycle of the system.

Taken together, space and time demonstrate that trustworthiness is not reducible to fixed properties or static compliance criteria. It is a relational attribute with the system, and it is contingent on the interaction between the system and its ever-changing socio-technical environment. Any meaningful attempt to design for or assess trustworthiness must therefore account for its spatial embeddedness and temporal evolution.

VI. IMPACT OF TRUSTWORTHINESS IN REQUIREMENT ENGINEERING

In this section, we focus on discussing the impact of trustworthiness in requirement engineering.

Relationship between Quality and Trustworthiness – The work in [9] foresees a reconsideration of quality to include new dimensions related to human, societal, and environmental values. In the last few years, standardization bodies proposed more than 25 standards in AI to alleviate the lack of good practices for governance, management and quality of AI systems [26]. These standards focus on specific aspects, e.g., big data, data quality, risk, safety, security, governance, etc. There have been also a number of extensions to the ISO/IEC 25010:2023 [27] Systems and software Quality Requirements and Evaluation (SQuaRE) standards to meet the needs of quality of modern systems [28], [29]. This flourishing of standards and extensions easily creates confusion. To make standards truly useful, an activity of harmonization is needed, e.g., through the definition of an ontology to unify the terms used in the different standards (those from SC42, from SC7, and others) [26].

Overall, the traditional understanding of quality is challenged. For instance, the advent of LLMs for code generation or for supporting development could potentially

tune down the need for reuse or could change the concept of reuse we are used to. In fact, instead of producing reusable code that can be easily adapted and customized to the actual needs, LLMs can generate new pieces of code on demand. So, one open question is: reusability will lose importance in the near future? Similar reasoning can be done for other qualities like maintainability. On another perspective, new quality aspects are gaining increasing importance and they should be taken into account, like transparency, explainability, accountability, etc.

Elicitation and specification of new requirements – These new quality dimensions embedding individual preferences and even moral views raise questions about whether they should be classified as functional or non-functional requirements or whether such classification is still meaningful or enough [9]. It becomes also difficult to elicit stakeholders' needs. The complexity comes also from bringing together different communities, like philosophers, computer scientists, psychologists, towards new solutions that integrate concepts and research results from these communities.

Another challenge concerns the operationalization of new requirements so that they become machine readable and automatically and programmatically interpretable. Townsend et al. [30] introduced the concept of SLEEC (social, legal, ethical, empathetic, or cultural) rules and they also outline a methodology for eliciting them. Other works proposed approaches to systematically translate SLEEC rules into a formal language to support automated reasoning, thus enabling their effective use in AI systems (e.g., [31], [32]). Nevertheless, SLEEC rules seem to cover only a small part of trustworthiness. Additionally, the elicitation and specification of trustworthiness should take into account its dynamic nature, connected with the dependence on humans' subjectivity and its evolution over time. The work in [33], [34] investigates how to collect personal ethical preferences and their seamless and consistent combination with other relevant dimensions starting from a top-down ethical perspective. In summary, more effort is needed to understand how to specify each aspect of trustworthiness when we will have a clear understanding of it in each of its facets.

From tradeoff to runtime negotiation of requirements – The subjective dimension of trustworthiness is adding additional challenges. In scenarios of our daily life, we live in an environment that is shared with other humans and various technologies. Easily, subjective choices of the humans involved in a given scenario can lead to conflicting specifications. This calls for a tradeoff and/or negotiation of values among the involved humans and digital choices [35]. We could be tempted to treat these conflicting choices as traditional non-functional requirements like security and performance and their tradeoff, as done in [36]. Following what discussed in [9], we believe that the traditional tradeoff, which can be solved at design time, is not enough. First, in various scenarios we only identify the involved humans at

runtime and in a specific moment. Second, to find a solution that could satisfy the involved humans, we would probably need a proper negotiation, with potential relaxations and downgrading of specific choices. Moreover, the subjective, selective, context-aware - and, in some ways, opportunistic - attribution of trustworthiness reinforces the idea that requirements related to trust require flexible, runtime-capable systems that can adapt, negotiate, and calibrate their behavior in response to changing user needs, situational stakes, and perceived risks.

Requirements validation: V&V for trustworthiness – Systems are increasingly required to operate in partially known, structured, and controllable environments, and they are subject to various sources of uncertainty [37]. Then, it becomes necessary and unavoidable to build guarantees of correct behavior of systems at runtime time [38]. This calls for runtime verification techniques [39], [40] and field-based testing techniques [41], [42]. Despite the various progresses of the community, there is the need of techniques able to provide evidence of assurance for the various dimensions of trustworthiness like auditability, accountability, fairness, non-discrimination, transparency, etc.

VII. CONCLUSION AND FUTURE WORKS

In this paper, we delved into the concept of trustworthiness in AI-powered systems, starting from its conceptual complexity and definition ambiguities, and then investigating related practical implications for requirement engineering. Based on the distinction between trustworthiness and reliance, we argued that the former cannot be reduced to technical reliability, especially in a context where AI systems are increasingly autonomous and, moreover, are strongly embedded in our social and private lives. Drawing on recent philosophical and technical literature on the discourse, we argued that trustworthiness must be understood - and then operationalized - as a multi-dimensional, dynamic, contextual construct, emerging at the intersection of technical performance, ethical alignment with human values and policies, and also user perception and expectations.

For this reason, we outlined a first path to reach conceptual clarity, defining three forms of trustworthiness - technical, ethical, and perceived - each raising distinct challenges for the elicitation, operationalization and validation of requirements. In this view, we demonstrated how trustworthiness is far from being a single requirement to be checked, but rather a nuanced property that depends on the satisfaction of diverse, transversal, and often interacting factors and requirements. We therefore came to the conclusion that such structural ambiguity - that sees trustworthiness sometimes as a mean, others as an end - complicates both the design and the interaction experience, the development and the assessment.

Indeed, from the perspective of requirement engineering, we discussed how this complexity impacts the processes of requirements elicitation, specifications, tradeoff analyses, and validation. In this sense, the dynamic, context-sensitive, and relational nature of trustworthiness make it a slippery notion,

that calls for new aware interdisciplinary methods to bridge the gap between philosophical discourse (e.g., value alignment, and machine's moral load) and technical needs (e.g., runtime verification processes and SLEEC values elicitation frameworks).

As for the future development of this work, our aim is to develop a proper framework to support developers, designers, validation teams, and other practitioners in truly grasping the layered nature of trustworthiness in AI systems. This framework will offer real-world scenarios, illustrative examples, as well as a clear visual representation of the different dimensions involved. It will serve as a conceptual pipeline to guide the integration of trustworthiness throughout the AI system lifecycle — from conceptual clarity to design and deployment implications' awareness, from validation to continuous adaptation. In conclusion, we aim to provide not only a vocabulary for thinking about trustworthiness, but also a usable methodology for building AI-powered systems that are not only technically performative, but also ethically and socially meaningful.

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